

# Chapter 10d: Multifunction Integral & Inclusions

Nonlinear Analysis and Fixed Point Theory

Based on Pathak (2018)

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# Overview: Multifunctions and Inclusions

## Key Topics:

- 1 Set-valued mappings (multifunctions)
- 2 Measurability and integrability for multifunctions
- 3 Integrals of set-valued functions (Aumann integral)
- 4 Differential inclusions
- 5 Existence theorems

## Motivation:

- Real-world systems often involve uncertainty and set-valued data
- Optimal control: control values constrained to a set
- Differential inclusions unify ODE and discontinuous systems
- Applications in delay equations, stochastic systems

# Definition of Multifunction

## Definition (Multifunction)

Let  $X$  and  $Y$  be topological spaces. A **multifunction** (or **set-valued mapping**) is a map

$$F : X \rightarrow 2^Y$$

where  $2^Y$  denotes the power set of  $Y$ . That is, for each  $x \in X$ ,  $F(x)$  is a subset of  $Y$ .

**Alternative notation:**

$$F : X \rightrightarrows Y, \quad F(x) \subseteq Y \text{ for all } x \in X$$

**Key distinction from single-valued functions:**

- Single-valued:  $f : X \rightarrow Y$  assigns one element  $f(x) \in Y$
- Multifunction:  $F : X \rightrightarrows Y$  assigns a set  $F(x) \subseteq Y$

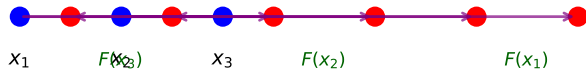
**Examples:**

- $F(x) = [0, x]$  for  $x \in \mathbb{R}$
- $F(x) = \{y \in Y : d(x, y) \leq r\}$  (ball-valued)
- $F(x) = \{y : g(x, y) \leq 0\}$  (constraint set)

# Visualizing Multifunctions

**Domain  $X$**

**Codomain  $Y$**



# Selections of Multifunctions

## Definition (Selection)

Let  $F : X \rightrightarrows Y$  be a multifunction. A function  $f : X \rightarrow Y$  is called a **selection** (or **branch**) of  $F$  if

$$f(x) \in F(x) \quad \text{for all } x \in X$$

**Notation:**  $f \in \mathcal{S}(F)$  denotes that  $f$  is a selection of  $F$ .

## Lemma (Existence of Measurable Selection)

If  $F : \Omega \rightarrow 2^{\mathbb{R}^n}$  is a multifunction with nonempty closed values and  $F$  has a measurable graph, then  $F$  admits a measurable selection  $f : \Omega \rightarrow \mathbb{R}^n$ .

**Domain**  $X = \mathbb{R}$

**Sets in Codomain**  $Y = \mathbb{R}$



# Basic Measurability Concepts

## Definition (Closed-valued Multifunction)

A multifunction  $F : \Omega \rightarrow 2^{\mathbb{R}^n}$  is **closed-valued** if  $F(\omega)$  is a closed subset of  $\mathbb{R}^n$  for each  $\omega \in \Omega$ .

## Definition (Graph of Multifunction)

The **graph** of a multifunction  $F : \Omega \rightarrow 2^{\mathbb{R}^n}$  is defined as

$$\text{Gr}(F) = \{(\omega, x) : \omega \in \Omega, x \in F(\omega)\}$$

## Definition ( $\mathcal{A}$ -measurability)

A closed-valued multifunction  $F : \Omega \rightarrow 2^{\mathbb{R}^n}$  is said to be  **$\mathcal{A}$ -measurable** if its graph  $\text{Gr}(F)$  is measurable in the product space  $\Omega \times \mathbb{R}^n$ .

# Types of Measurability

## Closed-valued Multifunction

$F(\omega)$  is closed for each  $\omega \in \Omega$



Graph:  $\text{Gr}(F) = \{(\omega, x) : x \in F(\omega)\}$   
is closed in  $\Omega \times X$

## Measurability Types

### Strong Measurability

$F$  is measurable (preimage of open sets)

### Weak Measurability

Selections are measurable

### $\mathcal{A}$ -measurability

Closed graph and measurable projections

## Hierarchy of measurability:

# Measurable Selection Theorem

## Theorem (Measurable Selection - Kuratowski and Ryll-Nardzewski)

Let  $(\Omega, \mathcal{A}, \mu)$  be a measure space and  $F : \Omega \rightarrow 2^{\mathbb{R}^n}$  a multifunction with:

- Nonempty compact values:  $F(\omega) \neq \emptyset$  is compact for all  $\omega$
- Closed graph in the product space

Then  $F$  admits a measurable selection  $f : \Omega \rightarrow \mathbb{R}^n$  such that  $f(\omega) \in F(\omega)$  a.e.

### Proof Idea:

- 1 Fix countable dense set  $\{x_1, x_2, \dots\}$  in  $\mathbb{R}^n$
- 2 For each  $k$ , define  $A_k = \{\omega : x_k \in F(\omega)\}$  (measurable by closed graph)
- 3 Inductively select  $x_k$  to build a sequence converging to a selection

# Aumann Integral

## Definition (Aumann Integral)

Let  $(\Omega, \mathcal{A}, \mu)$  be a measure space and  $F : \Omega \rightarrow 2^X$  a multifunction into a Banach space  $X$ . The **Aumann integral** is defined as

$$\int_{\Omega} F(\omega) d\mu(\omega) = \left\{ \int_{\Omega} f(\omega) d\mu(\omega) : f \in S(F), f \text{ integrable} \right\}$$

where  $S(F)$  is the set of all integrable selections of  $F$ .

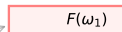
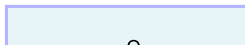
### Key features:

- The integral is a set (closure of convex hull of integrals of selections)
- Aggregates information from all possible selections
- Well-defined for convex-valued multifunctions

Parameter space  $\Omega$

Multifunction  $F : \Omega \rightarrow 2^X$  (closed, convex-valued)

Integral  $\int_{\Omega} F(\omega) d\mu(\omega)$



# Properties of the Aumann Integral

## Theorem (Basic Properties of Aumann Integral)

If  $F, G : \Omega \rightarrow 2^X$  are integrable multifunctions with convex compact values, then:

- ① (Nonempty):  $\int_{\Omega} F(\omega) d\mu(\omega) \neq \emptyset$
- ② (Closure): The integral is closed and convex
- ③ (Linearity): For  $A \in \mathcal{A}$ ,

$$\int_A (cF)(\omega) d\mu(\omega) = c \int_A F(\omega) d\mu(\omega), \quad c \geq 0$$

- ④ (Additivity):

$$\int_{\Omega} (F + G)(\omega) d\mu(\omega) = \int_{\Omega} F(\omega) d\mu(\omega) + \int_{\Omega} G(\omega) d\mu(\omega)$$

- ⑤ (Monotonicity): If  $F(\omega) \subseteq G(\omega)$  a.e., then

$$\int_{\Omega} F(\omega) d\mu(\omega) \subseteq \int_{\Omega} G(\omega) d\mu(\omega)$$

# Bochner Integral for Multifunctions

## Definition (Bochner Integrability)

Let  $X$  be a Banach space, and  $F : \Omega \rightarrow 2^X$  a multifunction. We say  $F$  is **Bochner integrable** if:

- 1  $F$  admits at least one measurable selection
- 2 There exists a measurable selection  $f$  such that  $\int_{\Omega} \|f(\omega)\|_X d\mu(\omega) < \infty$

## Proposition (Bochner Integral Representation)

If  $F : \Omega \rightarrow 2^X$  is Bochner integrable with convex values, then

$$\int_{\Omega} F(\omega) d\mu(\omega) = \text{cl} \left( \overline{\text{co}} \left\{ \int_{\Omega} f(\omega) d\mu(\omega) : f \in S(F) \right\} \right)$$

where  $\text{cl}$  denotes closure and  $\overline{\text{co}}$  denotes convex hull.

# Example: Computing Aumann Integral

**Problem:** Let  $\Omega = [0, 1]$  with Lebesgue measure, and  $F(\omega) = [-\omega, \omega]$ .

**Solution:**

All selections of  $F$  have the form  $f(\omega) = c(\omega)\omega$  where  $c(\omega) \in [-1, 1]$ .

```
import numpy as np
from scipy.integrate import quad

# Parametrize by c(omega) in [-1, 1]
def selection(omega, c):
    return c * omega

# Aumann integral is the set of integrals
# Integral of f(omega) = c*omega over [0,1] is c/2

# For c in [-1, 1], integral is in [-1/2, 1/2]
c_values = np.linspace(-1, 1, 101)
integrals = c_values / 2

# Aumann integral
min_integral = integrals.min() # -0.5
max_integral = integrals.max() # 0.5

print(f"Aumann integral of F: [{min_integral}, {max_integral}])")
```

**Result:**  $\int_0^1 F(\omega) d\omega = [-\frac{1}{2}, \frac{1}{2}]$

# Introduction to Differential Inclusions

## Definition (Differential Inclusion)

A **differential inclusion** is a problem of the form

$$\frac{dx}{dt} \in F(t, x(t)), \quad x(t_0) = x_0$$

where  $F : [t_0, T] \times \mathbb{R}^n \rightarrow 2^{\mathbb{R}^n}$  is a multifunction.

## Relationship to ODE:

- ODE:  $\frac{dx}{dt} = f(t, x)$  (single velocity)
- DI:  $\frac{dx}{dt} \in F(t, x)$  (set of possible velocities)

A **solution** is an **absolutely continuous function**  $x : [t_0, T] \rightarrow \mathbb{R}^n$  such that:

- 1  $x(t_0) = x_0$
- 2  $x'(t) \in F(t, x(t))$  for almost all  $t \in [t_0, T]$

# Examples of Differential Inclusions

## Example 1: Set-valued Lipschitz perturbations

$$\frac{dx}{dt} = f(t, x) + u(t), \quad u(t) \in U(t)$$

where  $U(t)$  is the constraint set (e.g., control constraints in optimal control).

## Example 2: Discontinuous systems

$$\frac{dx}{dt} \in \overline{\text{co}} \left\{ \lim_{y \rightarrow x, y \notin S} f(t, y) \right\}$$

where  $S$  is the set of discontinuities.

## Example 3: Delay differential equations

$$\frac{dx}{dt} \in F(t, x(t), x(t - \tau))$$

with distributed delays.

## Example 4: Control systems

$$\frac{dx}{dt} = A(t)x + B(t)u, \quad u(t) \in U, \quad x(0) = x_0$$

# Solution Concepts

## Definition (Carathéodory Solution)

An absolutely continuous function  $x : [t_0, T] \rightarrow \mathbb{R}^n$  is a **Carathéodory solution** of

$$\frac{dx}{dt} \in F(t, x), \quad x(t_0) = x_0$$

if:

- ①  $x(t_0) = x_0$
- ②  $x'(t) \in F(t, x(t))$  for almost all  $t \in [t_0, T]$

## Definition (Solution Set)

The **solution set** is

$$\mathcal{S}(F, x_0) = \{x : [t_0, T] \rightarrow \mathbb{R}^n \mid x \text{ solves the DI with initial condition } x_0\}$$

**Key question:** When is the solution set nonempty? When is it compact? Connected?

# Existence Theorem for Differential Inclusions

## Theorem (Existence - Convex Case)

Let  $F : [t_0, T] \times \mathbb{R}^n \rightarrow 2^{\mathbb{R}^n}$  satisfy:

- ① (Nonempty):  $F(t, x) \neq \emptyset$  for all  $(t, x)$
- ② (Convexity):  $F(t, x)$  is convex for all  $(t, x)$
- ③ (Measurability):  $F$  is  $\mathcal{A}$ -measurable in  $t$  for each fixed  $x$
- ④ (Continuity):  $F$  is upper semicontinuous in  $x$  for almost all  $t$
- ⑤ (Growth):  $\|F(t, x)\| \leq g(t)(1 + \|x\|)$  where  $g \in L^1([t_0, T])$

Then for any initial condition  $x_0 \in \mathbb{R}^n$ , there exists at least one solution to

$$\frac{dx}{dt} \in F(t, x), \quad x(t_0) = x_0$$

on  $[t_0, T]$ .

# Existence Theorem - Non-convex Case

## Theorem (Existence - Nonconvex Case)

Let  $F : [t_0, T] \times \mathbb{R}^n \rightarrow 2^{\mathbb{R}^n}$  satisfy:

- ① (Nonempty compact values):  $F(t, x)$  is compact for all  $(t, x)$
- ② (Closed graph):  $\text{Gr}(F)$  is closed in  $[t_0, T] \times \mathbb{R}^n \times \mathbb{R}^n$
- ③ (Measurable): For each fixed  $x$ ,  $F(\cdot, x)$  is measurable
- ④ (Growth):  $|F(t, x)| \leq g(t)(1 + |x|)$  where  $g \in L^1$

Then there exists at least one solution to  $\frac{dx}{dt} \in F(t, x)$ ,  $x(t_0) = x_0$ .

## Proof Strategy:

- ① Apply measurable selection theorem to extract a measurable selection
- ② Use growth condition to bound solutions
- ③ Apply Arzelà-Ascoli to extract convergent subsequence

# Uniqueness and Continuous Dependence

## Theorem (Uniqueness)

Under the conditions of the existence theorem, if additionally  $F$  satisfies:

- ① (Singleton):  $F(t, x)$  is a singleton, or
- ② (Lipschitz):  $F$  is Lipschitz continuous in  $x$  uniformly in  $t$

Then the solution is unique.

## Proposition (Continuous Dependence)

Let  $F_n \rightarrow F$  (in an appropriate sense) and  $x_{0,n} \rightarrow x_0$ . Then the solutions  $x_n$  of

$$\frac{dx_n}{dt} \in F_n(t, x_n), \quad x_n(0) = x_{0,n}$$

converge (in an appropriate norm) to the solution  $x$  of

$$\frac{dx}{dt} \in F(t, x), \quad x(0) = x_0$$

## Existence Theorems for Inclusions

**Condition 1: Compactness**  $F$  is compact domain  $\Omega$  or compact feasible set

**Condition 2: Convexity**  $F(\omega)$  is convex for all  $\omega$

**Condition 3: Measurability**  $F$  has measurable selections

**Condition 4: Growth condition**  $F(\omega)$  with  $g$  integrable

## Main Theorem (Existence Result)

If conditions 1-4 hold, then the inclusion

$$\frac{dx}{dt} \in F(t, x(t))$$

has at least one solution

# Relaxed Differential Inclusions

## Definition (Relaxed Inclusion)

Given a differential inclusion  $\frac{dx}{dt} \in F(t, x)$ , the **relaxed inclusion** is

$$\frac{dx}{dt} \in \overline{\text{co}}F(t, x)$$

where  $\overline{\text{co}}$  denotes the closure of the convex hull.

### Importance:

- The relaxed inclusion always has more solutions (includes all solutions of original)
- Often easier to analyze (convex-valued)
- Relaxation does not change the solution set for convex-valued  $F$
- Important for control theory and approximation theory

## Theorem (Relaxation Theorem)

If  $x$  is a solution of the relaxed inclusion  $\frac{dx}{dt} \in \overline{\text{co}}F(t, x)$ , then for any  $\epsilon > 0$ , there exists a solution  $x_\epsilon$  of the original inclusion  $\frac{dx}{dt} \in F(t, x)$  such that  $\|x(t) - x_\epsilon(t)\| < \epsilon$  for all  $t$ .

# Multivalued Operators and Accretivity

## Definition (Accretive Multifunction)

A multifunction  $A : \mathbb{R}^n \rightarrow 2^{\mathbb{R}^n}$  is **accretive** if for all  $x, y \in D(A)$  and  $u \in A(x)$ ,  $v \in A(y)$ ,

$$\langle u - v, x - y \rangle \geq -\omega \|x - y\|^2$$

for some constant  $\omega \geq 0$ .

## Proposition (Connection to Differential Inclusions)

The differential inclusion  $\frac{dx}{dt} \in A(x)$  where  $A$  is accretive admits solutions for any initial condition and has unique solutions under additional conditions (e.g.,  $A$  is maximal accretive).

### Applications:

- Subdifferential of convex functions
- Evolution equations in Banach spaces
- Nonlinear semigroups and abstract evolution

# Numerical Example: Discrete Approximation

**Problem:** Solve  $\frac{dx}{dt} \in F(t, x)$  where  $F(t, x) = [-1 - t, 1 + t] \cdot x$  on  $[0, 1]$ ,  $x(0) = 1$ .

```
import numpy as np
import matplotlib.pyplot as plt

# Discrete approximation (Euler method)
def solve_inclusion_bounds(T, n_steps):
    """Compute bounds on solution set"""
    t = np.linspace(0, T, n_steps)
    dt = t[1] - t[0]

    # Upper and lower bounds
    x_min = np.zeros(n_steps)
    x_max = np.zeros(n_steps)
    x_min[0] = x_max[0] = 1.0

    for i in range(n_steps - 1):
        t_i = t[i]
        # F(t, x) = [-(1+t)*x, (1+t)*x]
        # Min: use lower velocity
        x_min[i+1] = x_min[i] * np.exp(-(1+t_i)*dt)
        # Max: use upper velocity
        x_max[i+1] = x_max[i] * np.exp((1+t_i)*dt)

    return t, x_min, x_max
```

```
# Compute solution bounds
t, x_min, x_max = solve_inclusion_bounds(1.0, 100)
```

# Key Concepts Summary

- ① **Multifunctions:** Set-valued generalizations of single-valued functions
- ② **Measurability:** Multiple notions ( $\mathcal{A}$ -measurable, graph-measurable)
- ③ **Selections:** Single-valued approximations of multifunctions
- ④ **Aumann Integral:** Set integral as convex hull of integrals of selections
- ⑤ **Differential Inclusions:**  $\frac{dx}{dt} \in F(t, x)$  generalizes ODE
- ⑥ **Existence Theorems:** Under convexity + growth + measurability conditions
- ⑦ **Applications:** Control theory, delay equations, nonlinear PDE

# Main Results (Theorems)

- **Kuratowski-Ryll-Nardzewski:** Measurable selection exists for closed-valued multifunctions
- **Properties of Aumann Integral:** Closed convex sets, linearity, additivity
- **Existence for Convex DI:** Nonempty solution set under standard assumptions
- **Existence for Nonconvex DI:** Uses measurable selections + Arzelà-Ascoli
- **Relaxation Theorem:** Relaxed solutions approximate original solutions

# Recommended Reading

## Primary References:

- ① Pathak, H. K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*, Chapter 10
- ② Aubin, J. P., & Cellina, A. (1984). *Differential Inclusions: Set-Valued Maps and Viability Theory*
- ③ Kisielewicz, M. (1991). *Differential Inclusions and Optimal Control*

## Topics for Further Study:

- Viability theory and state constraints
- Optimal control of differential inclusions
- Measurable multifunctions and weak integrability
- Nonconvex analysis and nonsmooth optimization
- Stochastic differential inclusions

# Practice Problems

- 1 Show that if  $F(x) = [a(x), b(x)]$  where  $a, b$  are continuous, then  $F$  is upper semicontinuous.
- 2 Compute the Aumann integral  $\int_0^1 F(t) dt$  where  $F(t) = [-t, t^2]$ .
- 3 Verify that the convex hull of a multifunction  $\overline{\text{co}}F$  satisfies all properties of the Aumann integral.
- 4 Analyze the solution set of  $\frac{dx}{dt} \in \{-1, 0, 1\}x$  with  $x(0) = 1$ . Is it connected?
- 5 Prove that if  $F$  is Lipschitz continuous, then the solution of the DI is unique.

# Final Remarks

## Philosophical perspective:

- Multifunctions allow us to model systems with uncertainty, incompleteness, or nonuniqueness
- DI framework is more general than ODEs but still tractable for analysis
- Integration theory for multifunctions extends classical measure theory naturally

## Practical importance:

- Essential tool in optimal control, especially with constraints
- Handles discontinuities and nonsmooth phenomena
- Bridges classical analysis and set-valued topology
- Foundation for modern nonlinear analysis

Questions?